

lec-08

H-A-calculus-see-(13.1)

* A function of two variables, x and y , is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in the xy plane.

* A function f of three variables, x , y and z , is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in the three dimensional space.

* Example: let $f(x, y) = \sqrt{y+1} + \ln(x^2 - y^2)$.
Find $f(e, 0)$, $f(x, 0)$ and $f(e, y)$.

Ans:

$$f(x, y) = \sqrt{y+1} + \ln(x^2 - y^2)$$

$$\therefore f(e, 0) = \sqrt{0+1} + \ln(e^2 - 0^2)$$

$$= 1 + \ln e^2 = 1 + 2 \ln e = 1 + 2 = 3$$

$$f(x,0) = \sqrt{0+1} + \ln(x^x - 0^0) = 1 + 2 \ln x = h(x) \text{ (say)}$$

$$\Sigma f(e,y) = \sqrt{y+1} + \ln(e^x - y^y) = g(y) \text{ (say)}$$

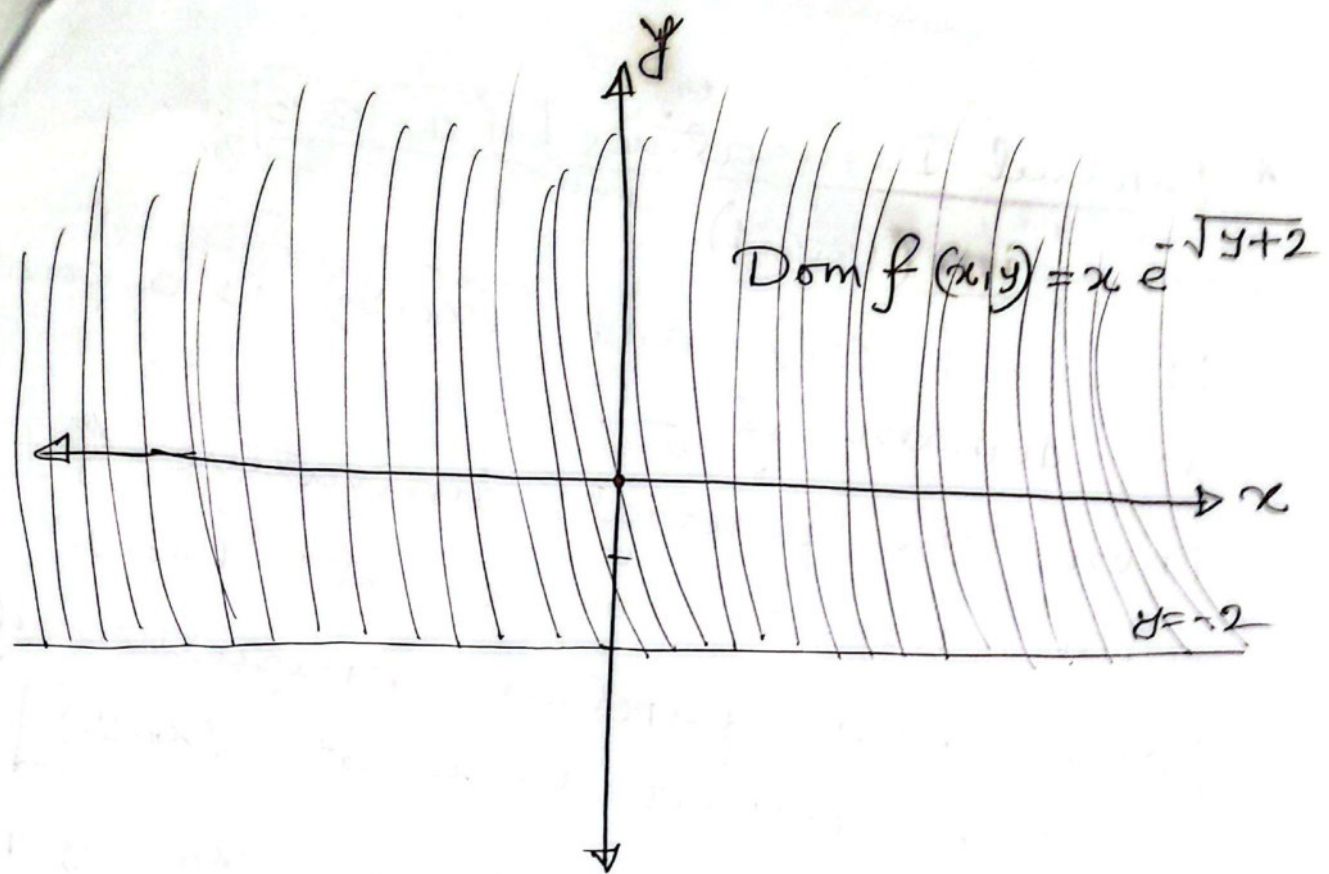
* N.B: For any fixed value of x then the function depends only on one variable y . and also for fixed value of y the function depends only on one variable x . ex. $h(x)$ & $g(y)$.

* Find the domain of the function
 $f(x,y) = x e^{-\sqrt{y+2}}$

Ans: Given that $f(x,y) = x e^{-\sqrt{y+2}}$

There is no value of x for which the function is undefined, so
 $x \in (-\infty, \infty)$.

For y , we set
 $y+2 \geq 0$



so the domain of the function is all points above or on the line $y = -2$ in x - y plane.

In Mathematical notation

$$\text{Dom } f = \left\{ (x, y) \in \mathbb{R}^2 : -\infty < x < \infty \text{ \& } -2 \leq y < \infty \right\}$$

$$\text{or } = \left\{ (x, y) \in \mathbb{R}^2 : x \in (-\infty, \infty) \text{ \& } y \in [-2, \infty) \right\}$$

* Partial Derivatives [H.A 13.3]

* Definition (13.3.1)

If $z = f(x, y)$ and (x_0, y_0) is a point in domain of f ,

then the partial derivative of f with respect to x at (x_0, y_0) [also called partial derivative of z with respect to x at (x_0, y_0)]

is the derivative at x_0 of the function that results when $y = y_0$

is held fixed and x is allowed to vary. This partial derivative

is denoted by $f_x(x_0, y_0)$ and is given by

$$\frac{\partial f}{\partial x} = f_x(x, y) = \left. \frac{d}{dx} f(x, y) \right|_{x=x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

Similarly,

The partial derivative of f with respect to y at (x_0, y_0) is the

derivative at y_0 of the function that results when $x = x_0$ is held fixed and y is allowed to vary.

This partial derivative is denoted by $f_y(x_0, y_0)$ and is given by

$$\frac{\partial f}{\partial y} = f_y(x_0, y_0) = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}$$

* Partial derivative Notation:

If $z = f(x, y)$, then partial derivatives f_x and f_y are also denoted by the symbols,

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = f_y$$

For particular point, like (x_0, y_0)

$$f_x(x_0, y_0) = \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}$$

$$\& \quad f_y(x_0, y_0) = \left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \left. \frac{\partial z}{\partial y} \right|_{(x_0, y_0)}$$

* Example: Find $\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$ for

i) $z = f(x, y) = x^4 \sin(xy^3)$

ii) $z = f(x, y) = x^2 + xy + y^3$

Ans:

(i) $z = x^4 f(xy^3) = x^4 \sin(xy^3)$

$$\text{so, } \frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (x^4 \sin(xy^3))$$

$$= x^4 \frac{\partial}{\partial x} \sin(xy^3) + \sin(xy^3) \frac{\partial}{\partial x} x^4$$

$$= x^4 \cdot \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3$$

$$= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3)$$

$$\& \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} x^4 \sin(xy^3)$$

$$= x^4 \frac{\partial}{\partial y} \sin(xy^3)$$

$$= x^4 \cos(xy^3) \cdot 3xy^2$$

$$\therefore \frac{\partial z}{\partial y} = f_y = 3x^5 y^2 \cos(xy^3)$$

$$\text{ii) } z = x^2 + xy + y^3$$

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (x^2 + xy + y^3)$$

$$= 2x + y + 0$$

$$\therefore \frac{\partial z}{\partial x} = 2x + y$$

$$\hookrightarrow \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (x^2 + xy + y^3)$$

$$= 0 + x + 3y^2$$

$$\therefore \frac{\partial z}{\partial y} = x + 3y^2$$

* problem (01) Evaluate f_x and f_y for

$$\text{a) } f(x, y) = x^2 + xy + y^4$$

$$\text{b) } f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{c) } f(x, y) = 3x^3y^2 + 2y^2 + 4x$$

* See Figure 13.3.1 From H. Anton book to understand the partial derivatives graphically.

* partial derivatives of functions with more than two variables:

For a function $w = f(x, y, z)$ of three variables, the partial derivative of f with respect to x , $f_x(x, y, z)$, is calculated by holding y and z constant and differentiating with respect to x . For f_y , x and z are considered as constants, and for f_z , x and y are considered as constants.

* let $w = f(x, y, z) = x^3 y^2 z^4 + 2xy + z$

$$\text{then } \frac{\partial w}{\partial x} = \frac{\partial f}{\partial x} = f_x(x, y, z) = 3x^2 y^2 z^4 + 2y + 0$$

$$\frac{\partial w}{\partial y} = \frac{\partial f}{\partial y} = f_y(x, y, z) = 2x^3 y z^4 + 2x + 0$$

$$\frac{\partial w}{\partial z} = f_z(x, y, z) = 4x^3 y^2 z^3 + 0 + 1$$

$$\text{and } f_z(-1, 1, 2) = 4(-1)^3(1)^2(2)^3 + 1 = -31$$

* Higher order partial Derivatives:

suppose that f is a function of two variables x and y . i.e., $f(x, y)$

- Differentiating f twice with respect to x

$$\rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} f_x = f_{xx}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

- Differentiating f twice w.r.t y -

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} f_y = f_{yy}$$

- Differentiating f first w.r.t x and then w.r.t y $\rightarrow$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = f_{xy}$$

- Differentiating f w.r.t y and then w.r.t x $\rightarrow$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = f_{yx}$$

- Also, $\frac{\partial^3 f}{\partial x^2 \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \right) = f_{yxx}$

for

* problem (p2): Evaluate f_{xx} , f_{yy} , f_{xy} , f_{xxy} , f_{xyx} and f_{yx} for the function.

a) $f(x,y) = \sin(x-y)$;

b) $f(x,y) = x^2 y^3 e^{xy^2}$

c) $f(x,y) = e^{\sin(\sqrt{x+y})} \cdot \tan y$.

* Practice problems!

13.1 $\rightarrow$ 1-4, 27

13.3 $\rightarrow$ 1-10, 25-40, 81, 82, 83, 84

* Example: If $f(x,y,z) = \sin(3x+yz)$, then evaluate f_{xyz} and f_{xxyz} .

Ans:

$$f(x,y,z) = \sin(3x+yz)$$

$$f_x = 3 \cos(3x+yz)$$

$$f_{xy} = (f_x)_y = \frac{\partial}{\partial y} (3 \cos(3x+yz))$$

$$= -3 \sin(3x+yz) \cdot (0+z)$$

$$f_{xy} = -3z \sin(3x+yz)$$

$$\begin{aligned} f_{xyz} &= (f_{xy})_z = \frac{\partial}{\partial z} f_{xy} \\ &= \frac{\partial}{\partial z} (-3z \sin(3x+yz)) \\ &= -3z \frac{\partial}{\partial z} \sin(3x+yz) \\ &\quad + \sin(3x+yz) \frac{\partial}{\partial z} (-3z) \end{aligned}$$

$$\begin{aligned} &= -3z \cos(3x+yz) \cdot (0+y) \\ &\quad - 3 \sin(3x+yz) \end{aligned}$$

$$\begin{aligned} f_{xyz} &= -3yz \cos(3x+yz) - 3 \sin(3x+yz) \end{aligned}$$

For $f_{xxy}z$

$$f_x = 3 \cos(3x+yz)$$

$$f_{xx} = -9 \sin(3x+yz)$$

$$\begin{aligned} f_{xxy} &= \frac{\partial}{\partial y} f_{xx} \\ &= \frac{\partial}{\partial y} (-9 \sin(3x+yz)) \end{aligned}$$

$$\therefore f_{xy} = -9z \cos(3x+yz)$$

$$\text{and } f_{xy}z = \frac{\partial}{\partial z} f_{xy}$$

$$= \frac{\partial}{\partial z} \left\{ -9z \cos(3x+yz) \right\}$$

$$= -9 \left[z \frac{\partial}{\partial z} \cos(3x+yz) + \cos(3x+yz) \frac{\partial}{\partial z} z \right]$$

$$= -9 \left[zy \sin(3x+yz) + \cos(3x+yz) \right]$$

$$f_{xy}z = 9yz \sin(3x+yz) - 9 \cos(3x+yz)$$

Ans.

practice . H. Anton

13.3 $\rightarrow$ 37-45

~~13.5~~ $\rightarrow$ 46-47